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Dear Hiring Manager,

I am writing to apply for the Machine Learning Engineer II role at TD in Toronto. As a Technical Consultant at Forestpin, a financial forensics and audit-analytics software company, I designed and deployed ML and statistical pipelines for financial forensics, compliance monitoring, and risk and anomaly detection on large structured enterprise datasets, and built SQL and Python backend scoring services that ran in production to support analyst triage. That work gave me direct, hands-on exposure to the kind of risk-scoring and compliance-monitoring problems a bank's ML team owns every day.

Since then, my PhD and postdoctoral work at the University of Trento have sharpened a discipline in rigorous model evaluation that I see as directly relevant to validating fraud and risk models before they reach production. On an EU Horizon-funded project, I designed frozen-protocol benchmark suites and ablation studies that quantified accuracy, latency, and CPU trade-offs for a real-time inference system, including a full written record of negative results, and I owned that pipeline end-to-end, from data acquisition through deployment and monitoring. Alongside this, I bring general ML engineering depth in Python (TensorFlow, PyTorch, scikit-learn), SQL, and production systems, as well as a systems-engineering discipline from five and a half years at MAS Holdings building embedded C++/Python inference and IoT pipelines for manufacturing at scale, including introducing CI/CD and code review practices.

I want to be direct about the shape of my background: my Forestpin role was six months and predates TD's scale of operation, so I have not worked with fraud or risk models at bank scale specifically. What I believe closes that gap is the combination of hands-on anomaly-detection experience, the benchmark and evaluation rigor from my research training, and the breadth of production ML engineering skills across data pipelines, model deployment, and systems programming that I have built since.

I am a Canadian Permanent Resident and eligible to work in Canada without sponsorship, based in Toronto, and available to start immediately. I would welcome the opportunity to discuss how this background could contribute to TD's ML engineering team.

Sincerely,
Nishal Stanislaus Silva